The Hessian Screening Rule

NeurIPS 2022 | New Orleans, USA

Johan Larsson[1,*] and Jonas Wallin[1]

[1]*The Department of Statistics, Lund University, Sweden*

[*]johan.larsson@stat.lu.se

Summary

We present a novel screening rule for the well-known lasso, using second-order information from the objective in order to improve the effectiveness of feature screening compared to competing methods.

Our method outperforms competing alternatives for a wide range of simulated and real data sets.

Problem

We focus on solving the lasso, which is represented by the following convex optimization problem,

$$\underset{\beta \in \mathbb{R}^p}{\text{minimize}} \left\{ f(\beta; X) + \lambda \|\beta\|_1 \right\},$$

along a regularization path $\lambda_1, \lambda_2 \ldots, \lambda_m$. (See Figure 1 for an example.)

The motivation for this focus is that the optimal λ is typically unknown and must be estimated through model tuning, such as cross-validation. This involves repeated refitting of the model to new batches of data, which is computationally demanding.

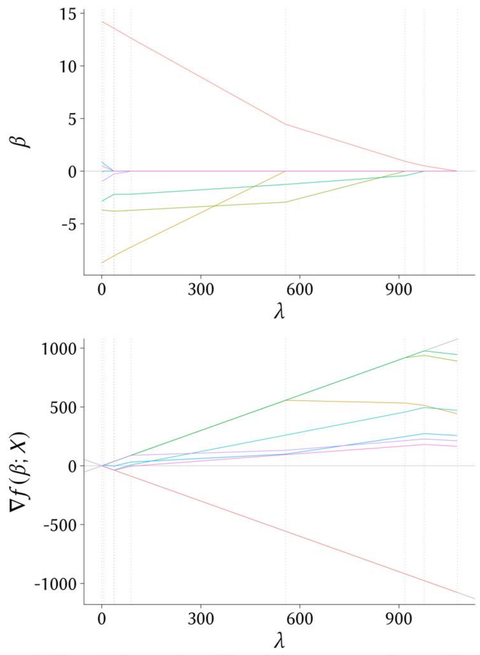

Figure 1: The solution vectors (β) and the corresponding gradients ($\nabla f(\beta; X)$) for a lasso regularization path. Note that the paths in both cases are piece-wise linear with kinks whenever a feature enters or leaves the model.

Screening Rules

Screening rules discard features before fitting the model. They work well because the solutions along the regularization path are sparse, particularly if $p \gg n$.

β is a solution to the lasso problem if it satisfies

$$\mathbf{0} \in \nabla f(\beta) + \lambda \partial \tag{1}$$

where ∂ is the subdifferential of $\|\beta\|_1$, defined as

$$\partial_j \in \begin{cases} \{\text{sign}(\beta_j)\} & \text{if } \beta_j \neq 0, \\ [-1, 1] & \text{otherwise.} \end{cases}$$

This means that $|\nabla f(\beta)_j| < \lambda$ necessitates that $\beta_j = 0$. But of course we do not have $\nabla f(\beta)$ prior to solving the problem. Instead, we replace it with an estimate.

Our Screening Rule

Our rule is based on the fact that we can express the solutions for the least-squares lasso along the path as a function of λ and the active set ($\mathcal{A}$):

$$\hat{\beta}(\lambda) = \left(X_{\mathcal{A}}{}^T X_{\mathcal{A}} \right)^{-1} \left(X_{\mathcal{A}}{}^T y - \lambda \, \text{sign}(\hat{\beta}_{\mathcal{A}}) \right).$$

This expression holds for an interval $[\lambda_k, \lambda_{k+1}]$ in which no changes occur in the active set, which means we can retrieve any solution in this range via

$$\hat{\beta}(\lambda_{k+1})_{\mathcal{A}} = \hat{\beta}(\lambda_k)_{\mathcal{A}} - (\lambda_k - \lambda_{k+1}) \left(X_{\mathcal{A}}{}^T X_{\mathcal{A}} \right)^{-1} \text{sign}\left(\hat{\beta}(\lambda_k)_{\mathcal{A}} \right).$$

Inserting this into the gradient gives us our estimate,

$$\nabla f\left(\hat{\beta}(\lambda_{k+1})_{\mathcal{A}} \right) = \nabla f\left(\hat{\beta}(\lambda_k)_{\mathcal{A}} \right) + (\lambda_k - \lambda_{k+1}) X^T X_{\mathcal{A}} \left(X_{\mathcal{A}}{}^T X_{\mathcal{A}} \right)^{-1} \text{sign}\left(\hat{\beta}(\lambda_k)_{\mathcal{A}} \right).$$

In Figure 2, we show a comparison of this estimate and the one used by the strong screening rule [1].

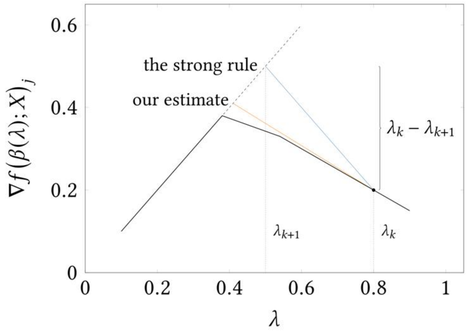

Figure 2: A comparison of the Hessian and strong screening rules. Our rule estimates the correlation of feature j with better precision and is hence less conservative.

Effectiveness

In Figure 3, we show the effectiveness of screening using our method compared to other screening rules. Our method is many times more effective than the contenders, especially for the high-correlation scenario.

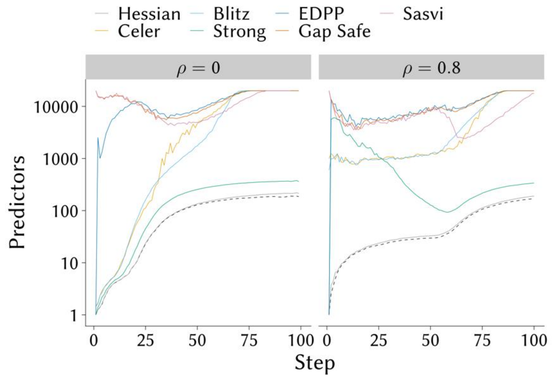

Figure 3: Number of features screened (included) when fitting a least-squares lasso path to a design with correlation ρ, $n = 200$, and $p = 20\,000$. The minimum number of active features at each step across iterations is given as a dashed line.

Simulated Data

In Figure 4, we show results of benchmarks comparing the performance of our algorithm, using the Hessian rule, against Celer [2], Blitz [3], and the working set strategy based on the strong rule [1]. Our method performs better than all of the competing methods, particularly when $n \gg p$ and correlation is high.

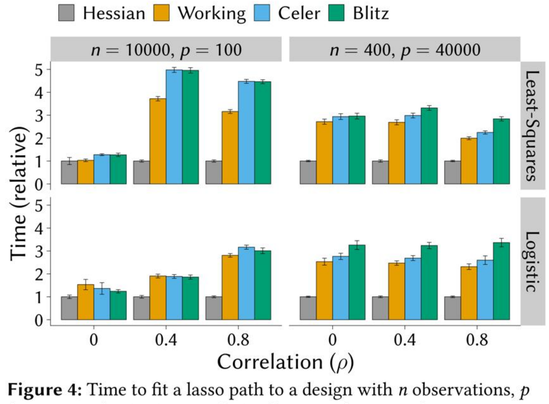

Figure 4: Time to fit a lasso path to a design with n observations, p features, and pairwise correlation between features of ρ.

Real Data

Our algorithm outperforms competing methods for real data too (Table 1). The algorithm is effective even in the case when $n \approx p$ and sparsity is high, which represents the worst-case scenario for the method.

Table 1: Average time to fit a lasso path to real data sets.

					Time (s)			
Data Set	n	p	Density	Loss	Hessian	Working	Blitz	Celer
bcTCGA	536	17 322	1	Least-Squares	3.0	7.7	12	11
e2006-log1p	16 087	4 272 227	1.4×10^{-3}	Least-Squares	200	440	760	840
e2006-tfidf	16 087	150 360	8.3×10^{-3}	Least-Squares	14	140	280	340
scheetz	120	18 975	1	Least-Squares	0.37	0.64	0.71	0.80
YearPredMSD	463 715	90	1	Least-Squares	79	540	710	710
arcene	100	10 000	5.4×10^{-1}	Logistic	4.4	3.3	4.4	4.0
colon-cancer	62	2000	1	Logistic	0.054	0.13	0.18	0.17
duke-cancer	44	7129	1	Logistic	0.11	0.21	0.25	0.26
ijcnn1	35 000	22	1	Logistic	0.94	5.5	4.7	3.5
madelon	2000	500	1	Logistic	48	230	240	250
news20	19 996	1 355 191	3.4×10^{-4}	Logistic	1300	1600	2200	2200
rcv1	20 242	47 236	1.6×10^{-3}	Logistic	130	270	380	380

Future Directions

Extending the screening rule to other methods such as SLOPE, MCP, and SCAD would be an interesting direction of research, as well as using the Hessian updates during optimization in order to speed up convergence.

References

[1] Robert Tibshirani et al. "Strong Rules for Discarding Predictors in Lasso-Type Problems". In: *Journal of the Royal Statistical Society. Series B: Statistical Methodology* 74.2 (Mar. 2012), pp. 245–266. ISSN: 1369-7412. DOI: 10/c4bb85.

[2] Mathurin Massias, Alexandre Gramfort, and Joseph Salmon. "Celer: A Fast Solver for the Lasso with Dual Extrapolation". In: *Proceedings of the 35th International Conference on Machine Learning*. ICML 2018. Ed. by Jennifer Dy and Andreas Krause. Vol. 80. Proceedings of Machine Learning Research. Stockholm, Sweden: PMLR, July 10–15, 2018, pp. 3315–3324.

[3] Tyler B Johnson and Carlos Guestrin. "Blitz: A Principled Meta-Algorithm for Scaling Sparse Optimization". In: *Proceedings of the 32nd International Conference on Machine Learning*. International Conference on Machine Learning. Vol. 37. Lille, France: JMLR: W&CP, 2015, p. 9.

Contact

- johan.larsson@stat.lu.se
- larssonjohan.com
- github.com/jolars
- fediscience.org/web/@jolars